1.

(a) Applying the equation for root-mean-square speed with the molar mass of hydrogen being 2.02 x 10^{-3} kg/mol, we have

$$v_{\text{rms}} = \sqrt{\frac{3(8.31\,\text{J/mol}\cdot\text{K})(4000\,\text{K})}{2.02\times10^{-3}\,\text{kg/mol}}} = 7.0\times10^{3}\,\text{m/s}.$$

(b) When the surfaces of the spheres that represent an H_2 molecule and an Ar atom are touching, the distance between their centers is the sum of their radius:

$$d = r_1 + r_2 = 0.5\times10^{-8}\,\text{cm} + 1.5\times10^{-8}\text{cm} = 2.0\times10^{-8}\,\text{cm}.$$

(c) The argon atoms are essentially at rest so in time t the hydrogen atom collides with all the argon atoms in a cylinder of radius d, and length vt, where v is its speed. That is, the number of collisions is $\pi d^2 vtN/V$, where N/V is the concentration of argon atoms. The number of collisions per unit time is

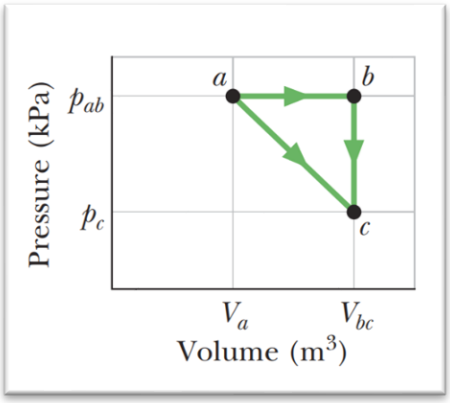
$$\frac{\pi d^2 vN}{V} = \pi\left(2.0\times10^{-10}\,\text{m}\right)^2\left(7.0\times10^{3}\,\text{m/s}\right)\left(4.0\times10^{25}\,\text{m}^{-3}\right) = 3.5\times10^{10}\,\text{collisions/s}.$$

2.

The internal energy is

$$E_{\text{int}} = \frac{3}{2}nRT = \frac{3}{2}(1.0\,\text{mol})(8.31\,\text{J/mol}\cdot\text{K})(273\,\text{K}) = 3.4\times10^{3}\,\text{J}.$$

3.



For any process that is depicted as a straight line on the pV diagram, the work is

$$W_{\text{straight}} = \left(\frac{p_i + p_f}{2}\right)\Delta V$$

which includes, as special cases, $W = p\Delta V$ for constant-pressure processes and $W = 0$ for constant-volume processes. Applying the equations for internal energy and degree of freedom, we have

$$E_{\text{int}} = n\left(\frac{f}{2}\right)RT = \left(\frac{f}{2}\right)pV$$

where we have used the ideal gas law in the last step. We emphasize that, in order to obtain work and energy in joules, pressure should be in pascals (N/m^2) and volume should be in cubic meters. The degrees of freedom for a diatomic gas is $f = 5$.

(a) The internal energy change is

$$E_{\text{int}\,c} - E_{\text{int}\,a} = \frac{5}{2}(p_c V_c - p_a V_a) = \frac{5}{2}\left((2.0\times10^3\ \text{Pa})(4.0\,\text{m}^3) - (5.0\times10^3\ \text{Pa})(2.0\,\text{m}^3)\right)$$
$$= -5.0\times10^3\ \text{J}.$$

(b) The work done during the process represented by the diagonal path is

$$W_{\text{diag}} = \left(\frac{p_a + p_c}{2}\right)(V_c - V_a) = (3.5\times10^3\ \text{Pa})(2.0\,\text{m}^3)$$

which yields $W_{\text{diag}} = 7.0\times10^3$ J. Consequently, the first law of thermodynamics gives

$$Q_{\text{diag}} = \Delta E_{\text{int}} + W_{\text{diag}} = (-5.0\times10^3 + 7.0\times10^3)\ \text{J} = 2.0\times10^3\ \text{J}.$$

(c) The fact that ΔE_{int} only depends on the initial and final states, and not on the details of the "path" between them, means we can write $\Delta E_{\text{int}} = \Delta E_{\text{int}\,c} - \Delta E_{\text{int}\,a} = -5.0\text{x}10^3$ J for the indirect path, too. In this case, the work done consists of that done during the constant pressure part (the horizontal line in the graph) plus that done during the constant volume part (the vertical line):

$$W_{\text{indirect}} = (5.0\times10^3\ \text{Pa})(2.0\,\text{m}^3) + 0 = 1.0\times10^4\ \text{J}.$$

Now, the first law of thermodynamics leads to

$$Q_{\text{indirect}} = \Delta E_{\text{int}} + W_{\text{indirect}} = (-5.0\times10^3 + 1.0\times10^4)\ \text{J} = 5.0\times10^3\ \text{J}.$$

4.

The molar specific heat at constant volume for a gas is given by as $C_V = \Delta E_{\text{int}} / n\Delta T$. Our system consists of three non-interacting gases.

When the temperature changes by ΔT the internal energy of the first gas changes by $n_1 C_{V1} \Delta T$, the internal energy of the second gas changes by $n_2 C_{V2} \Delta T$, and the internal energy of the third gas changes by $n_3 C_{V3} \Delta T$. The change in the internal energy of the composite gas is

$$\Delta E_{\text{int}} = (n_1 C_{V1} + n_2 C_{V2} + n_3 C_{V3})\,\Delta T.$$

This must be $(n_1 + n_2 + n_3)\, C_V \Delta T$, where C_V is the molar specific heat of the mixture.

Thus,

$$C_V = \frac{n_1 C_{V1} + n_2 C_{V2} + n_3 C_{V3}}{n_1 + n_2 + n_3}.$$

With n_1=2.40 mol, C_{V1}=12.0 J/mol·K for gas 1, n_2=1.50 mol, C_{V2}=12.8 J/mol·K for gas 2, and n_3=3.20 mol, C_{V3}=20.0 J/mol·K for gas 3, we obtain

$$C_V = \frac{(2.40\ \text{mol})(12.0\ \text{J/mol}\cdot\text{K}) + (1.50\ \text{mol})(12.8\ \text{J/mol}\cdot\text{K}) + (3.20\ \text{mol})(20.0\ \text{J/mol}\cdot\text{K})}{2.40\ \text{mol} + 1.50\ \text{mol} + 3.20\ \text{mol}}$$
$$= 15.8\ \text{J/mol}\cdot\text{K}$$

for the mixture.

5.

We have

$$\Delta E_{\text{int}} = nC_V\Delta T = \frac{5}{2}nR\Delta T, \quad Q = nC_p\Delta T = \frac{7}{2}nR\Delta T.$$

Dividing the equations, we obtain

$$\frac{\Delta E_{\text{int}}}{Q} = \frac{5}{7}.$$

Thus, the given value Q = 70 J leads to ΔE_{int} = 50 J.

6.

(a) With M = 32.0 g/mol, we have

$$n = \frac{M_{\text{sam}}}{M} = \frac{12.0\ \text{g}}{32.0\ \text{g/mol}} = 0.375\ \text{mol}.$$

(b) This is a constant pressure process with a diatomic gas. We note that a change of Kelvin temperature is numerically the same as a change of Celsius degrees.

$$Q = nC_p\Delta T = n\left(\frac{7}{2}R\right)\Delta T = (0.375\ \text{mol})\left(\frac{7}{2}\right)(8.31\ \text{J/mol}\cdot\text{K})(100\,\text{K}) = 1.09\times10^3\ \text{J}.$$

(c) From (b) we have:

$$\frac{\Delta E_{\text{int}}}{Q} = \frac{n\left(\frac{5}{2}R\right)\Delta T}{n\left(\frac{7}{2}R\right)\Delta T} = \frac{5}{7} \approx 0.714.$$

7.

(a) Assumed the gas to obey the ideal gas law, we use $V_f / V_i = \frac{1}{2}$.

$$p_iV_i^\gamma = p_fV_f^\gamma \Rightarrow \frac{p_f}{p_i} = \left(\frac{V_i}{V_f}\right)^\gamma = (2.00)^{1.3}$$

which yields p_f = (2.46)(1.0 atm) = 2.46 atm.

(b) Similarly,

$$T_f = T_i\left(\frac{V_i}{V_f}\right)^{\gamma-1} = (273\,\text{K})(1.23) = 336\,\text{K}.$$

(c) We use the gas law in ratio form and note that when p_1 = p_2 then the ratio of volumes is equal to the ratio of (absolute) temperatures. Consequently, with the

subscript 1 referring to the situation (of small volume, high pressure, and high temperature) the system is in at the end of part (a), we obtain

$$\frac{V_2}{V_1} = \frac{T_2}{T_1} = \frac{273\,\text{K}}{336\,\text{K}} = 0.813.$$

The volume V_1 is half the original volume of one liter, so

$$V_2 = 0.813(0.500\,\text{L}) = 0.406\,\text{L}.$$

8.

Since the process is adiabatic:

$p_i V_i^{\gamma} = p_f V_f^{\gamma}$.

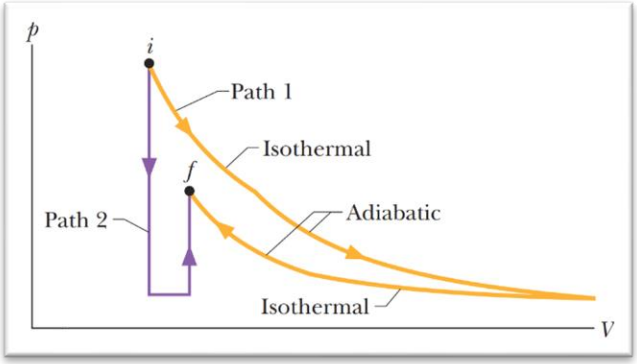
$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}$

$p_i V_i^{\gamma} = p_f V_f^{\gamma} \rightarrow (V_i/V_f)^{\gamma} = p_f/p_i = 5 \rightarrow V_i/V_f = \sqrt[\gamma]{5}$

$Tf = T_i(Vi/V_f)^{\gamma-1}$

With $\gamma = 4/3$, which gives $(1-\gamma)/\gamma = -1/4$, the temperature at the end of the adiabatic expansion is: 186K = -87°C

9.



Since ΔE_{int} does not depend on the type of process,

$$(\Delta E_{\text{int}})_{\text{path 2}} = (\Delta E_{\text{int}})_{\text{path 1}}.$$

Also, since (for an ideal gas) it only depends on the temperature variable (so ΔE_{int} = nCvΔT = 0 for

isotherms), then

$$(\Delta E_{\text{int}})_{\text{path1}} = \sum(\Delta E_{\text{int}})_{\text{adiabat}}.$$

Finally, since $Q = 0$ for adiabatic processes, then (for path 1)

$$(\Delta E_{\text{int}})_{\text{adiabatic expansion}} = -W = -40\ \text{J}$$
$$(\Delta E_{\text{int}})_{\text{adiabatic compression}} = -W = -(-25)\,\text{J} = 25\ \text{J}.$$

Therefore, $(\Delta E_{\text{int}})_{\text{path 2}}$ = -40J + 25J = -15J .

10.

Considering an adiabatic process, we have:

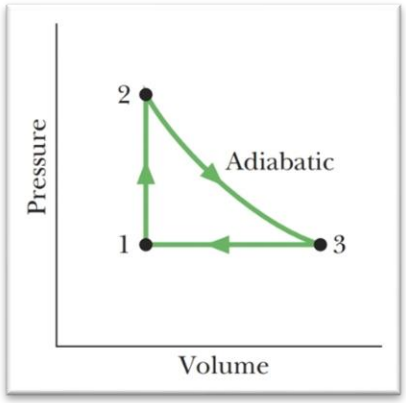
PV^{γ} = const → Vf = ….

→$PV^{\gamma} = P_iV_i^{\gamma}$ → $P = P_iV_i^{\gamma}/V$

$$W = p_iV_i^{\gamma}\int_{V_i}^{V_f} V^{-\gamma}dV = p_iV_i^{\gamma}\frac{V_f^{1-\gamma} - V_i^{1-\gamma}}{1-\gamma}.$$

We have $\gamma = 7/5$ and $P_f/P_i = 2$. Converting the initial pressure to pascals we find P_iV_i = 24240 J. Plugging in, then, we obtain W = -1.33 x 10^4 J.

11.



In the following, monatomic gas: $C_V = \frac{3}{2}R$ is the molar specific heat at constant volume, $C_P = \frac{5}{2}$R is the molar specific heat at constant pressure, ΔT is the temperature change, and n is the number of moles.

The process 1→2 takes place at constant volume.

(a) The heat added is

$$Q = nC_V\,\Delta T = \frac{3}{2}nR\,\Delta T = \frac{3}{2}(1.00\,\text{mol})(8.31\,\text{J/mol}\cdot\text{K})(600\,\text{K} - 300\,\text{K}) = 3.74\times10^3\ \text{J}.$$

(b) Since the process takes place at constant volume: W = 0, and the first law of thermodynamics tells us that the change in the internal energy is

$$\Delta E_{\text{int}} = Q = 3.74\times10^{3}\ \text{J}.$$

(c) The work W done by the gas is zero.

The process 2→3 is adiabatic.

(d) Q =0.

(e) The change in the internal energy is

$$\Delta E_{\text{int}} = nC_V\,\Delta T = \frac{3}{2}nR\,\Delta T = \frac{3}{2}(1.00\,\text{mol})(8.31\,\text{J/mol}\cdot\text{K})(455\,\text{K}-600\,\text{K}) = -1.81\times10^{3}\ \text{J}.$$

(f) According to the first law of thermodynamics the work done by the gas is

$$W = Q - \Delta E_{\text{int}} = +1.81\times10^{3}\ \text{J}.$$

The process 3→1 takes place at constant pressure.

(g) The heat added is

$$Q = nC_p\Delta T = \frac{5}{2}nR\Delta T = \frac{5}{2}(1.00\ \text{mol})(8.31\,\text{J/mol}\cdot\text{K})(300\,\text{K}-455\,\text{K}) = -3.22\times10^{3}\ \text{J}.$$

(h) The change in the internal energy is

$$\Delta E_{\text{int}} = nC_V\Delta T = \frac{3}{2}nR\Delta T = \frac{3}{2}(1.00\,\text{mol})(8.31\,\text{J/mol}\cdot\text{K})(300\,\text{K}-455\,\text{K}) = -1.93\times10^{3}\ \text{J}.$$

(i) According to the first law of thermodynamics the work done by the gas is

$$W = Q - \Delta E_{\text{int}} = -3.22\times10^{3}\ \text{J} + 1.93\times10^{3}\ \text{J} = -1.29\times10^{3}\ \text{J}.$$

(j) For the entire process the heat added is

$$Q = 3.74\times10^{3}\ \text{J} + 0 - 3.22\times10^{3}\ \text{J} = 520\ \text{J}.$$

(k) The change in the internal energy is

$$\Delta E_{\text{int}} = 3.74\times10^{3}\ \text{J} - 1.81\times10^{3}\ \text{J} - 1.93\times10^{3}\ \text{J} = 0.$$

(l) The work done by the gas is

$$W = 0 + 1.81\times10^{3}\ \text{J} - 1.29\times10^{3}\ \text{J} = 520\ \text{J}.$$

(m) We first find the initial volume. Use the ideal gas law $p_1V_1 = nRT_1$ to obtain

$$V_1 = \frac{nRT_1}{p_1} = \frac{(1.00\,\text{mol})(8.31\,\text{J/mol}\cdot\text{K})(300\,\text{K})}{(1.013\times10^{5}\ \text{Pa})} = 2.46\times10^{-2}\ \text{m}^3.$$

(n) Since 1→2 is a constant volume process, $V_2 = V_1 = 2.46 \times 10^{-2}\ \text{m}^3$. The pressure for state 2 is

$$p_2 = \frac{nRT_2}{V_2} = \frac{(1.00\ \text{mol})(8.31\,\text{J/mol}\cdot\text{K})(600\,\text{K})}{2.46\times10^{-2}\,\text{m}^3} = 2.02\times10^{5}\,\text{Pa}.$$

This is approximately equal to 2.00 atm.

(o) 3→1 is a constant pressure process. The volume for state 3 is

$$V_3 = \frac{nRT_3}{p_3} = \frac{(1.00\,\text{mol})(8.31\,\text{J/mol}\cdot\text{K})(455\,\text{K})}{1.013\times10^5\,\text{Pa}} = 3.73\times10^{-2}\,\text{m}^3.$$

(p) The pressure for state 3 is the same as the pressure for state 1: $p_3 = p_1$ = 1.013 x 10^5 Pa (1.00 atm)

MCQs

1. For an ideal diatomic gas:

a. C_v = 20.8 J/mol.K
b. C_v = 28.5 J/mol.K
c. C_v = 20.8 J/K
d. C_v = 25.5 J/K

ANSWER: A

2. A diatomic gas, compressed isobarically, did 42 J of work. Which was the amount of heat given to the gas during such process?

a. 147 J
b. 137 J
c. 127 J
d. 117 J

ANSWER: A

6. For a diatomic gas, the value of $\gamma = \frac{C_p}{C_v}$ is:

a. $\frac{9}{7}$
b. $\frac{7}{5}$
c. $\frac{5}{3}$
d. $\frac{5}{7}$

ANSWER: B

3. 1.80 mol of an ideal gas is taken from a volume of 3.00 m^3 to a volume of 1.50 m^3 via an isothermal compression at 30°C. How much energy is transferred as heat during the compression?

a. – 3140 J
b. – 3.14 J
c. – 314 J
d. – 31.4 J

ANSWER: A

4. The adiabatic relation between volume and temperature of an ideal gas is:

a. $V^{\gamma-1}\,T$ = constant
b. $V^{1+\gamma}\,T^{\gamma}$ = constant
c. $V^{1-\gamma}\,T^{\gamma+1}$ = constant
d. $V^{1-\gamma}\,T^{\gamma-1}$ = constant

ANSWER: A

5. Which of the following statements is correct

a. C_p can never be less than C_v
b. C_p C_v in some specific cases
c. C_p may be less than C_v depending on the gas
d. C_p may be equal to or less than C_v

ANSWER: A

7. The work done by an adiabatic expansion is written as:

a. $W = 0$
b. $W = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$
c. $W = P_2V_2 \ln(\frac{P_1}{P_2})$
d. $W = \frac{P_2V_2 - P_1V_1}{1-\gamma}$

ANSWER: D

$pV^{\gamma} = constant$ for adiabatic process → p = K/V^{γ} where K = $p_1{V_1}^{\gamma} = p_2{V_2}^{\gamma}$

$W = \int_{V_1}^{V_2} pdV = \int_{V_1}^{V_2} \frac{K}{V^{\gamma}} dV = \frac{K}{1-\gamma}\,(\frac{1}{{V_2}^{\gamma-1}} - \frac{1}{{V_1}^{\gamma-1}}) = \frac{1}{1-\gamma}\,(\frac{p_2{V_2}^{\gamma}}{{V_2}^{\gamma-1}} - \frac{p_1{V_1}^{\gamma}}{{V_1}^{\gamma-1}}) = \frac{P_2V_2 - P_1V_1}{1-\gamma})$

8. A monatomic gas is suddenly compressed to $(\frac{1}{8})^{th}$ of its initial volume adiabatically. The ratio of its final pressure to the initial pressure is

a. 32
b. 8
c. $\frac{1}{32}$
d. $\frac{1}{8}$

Answer: A

$p_1{V_1}^{\gamma} = p_2{V_2}^{\gamma}$ → $\frac{P1}{P2} = (\frac{1}{8})^{\gamma}$, with $\gamma = \frac{5}{3} for\ a\ monatomic\ gas$→ P1/P2 = 32

9. One mole of a monoatomic ideal gas is mixed with one mole of a diatomic ideal gas. The molar specific heat of the mixture at constant volume is:

a. R
b. $2R$
c. 0.5 R
d. 4R

Answer: B

Cv = $\frac{n1Cv1 + n2Cv2}{n1+n2} = \frac{1*\frac{3}{2}R+1*\frac{5}{2}R}{2} = 2R$

10. A gaseous mixture consists of 16g of Helium and 16g of Oxygen. The ratio $\frac{C_p}{C_v}$ of the mixture is:

a. 1.4
b. 1.54
c. 1.59
d. 1.62

Answer: D

Cv = $\frac{n1Cv1 + n2Cv2}{n1+n2}$

nHe = 16/4 = 4 mol, nO2 = 16/32 = 0.5 mol, Cv(He) = 3/2R, Cv(O2) = 5/2R